

Varun Manjunath

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RESEARCH INTERESTS

- Efficient LLM Serving Systems
 - Wafer-Scale Inference
- Energy-Efficient DNN Inference
 - Mixed-Precision Inference
 - Sparsity (Bit Level & Operand Level)
- Neuromorphic Computing
 - Spiking Neural Networks

RESEARCH SUMMARY

My research builds hardware and systems for **efficient machine learning inference**, from **energy-constrained edge accelerators** to **wafer-scale LLM serving**. Across these settings, I study **workload asymmetry** at three levels of granularity, **bit, operand and request**: work that looks uniform to the hardware varies widely in cost, so the slowest unit dictates both latency and utilization. I address this asymmetry through **hardware–software co-design** that exposes work at the finest schedulable granularity and balances it at runtime. My work has appeared at **MICRO, ICCAD and ASP-DAC**.

EDUCATION

Ph.D., Electrical and Computer Engineering, University of Southern California

Jan. 2026 — Present

Advisor: Prof. Priyadarshini Panda

Expected: May. 2030

B.Tech (Honours) + M.Tech, Electrical Engineering, Indian Institute of Technology Madras

Aug. 2020 — May. 2025

Advisor: Prof. Gopalakrishnan Srinivasan

GPA: 9/10

ON GOING WORK

FUSE: Fine-Grained Unified Stage Execution for Multimodal LLM Serving on Wafer-Scale Chips.

- **Problem:** Wafer-scale multimodal LLM serving fixes encode/prefill/decode partitions offline, which break as request mixes shift, while GPU-era schedulers allocate whole devices and so cannot exploit the per-core allocation and die-to-die bandwidth that wafer-scale chips offer
- **Solution:** An online, deadline-driven orchestrator that sizes and places every dispatch at runtime; $2.31\times$ lower mean TTFT ($4.23\times$ at p90), $1.52\times$ lower TPOT and $1.34\times$ goodput over a tuned disaggregated baseline, and $7.9\times$ lower TPOT and $3.0\times$ lower end-to-end latency than vLLM on 16 A100s

PUBLICATIONS [SELECTED]

BRIM: Workload-Balanced Dual-Sided Bit-Serial Sparse Inference Accelerator

Varun Manjunath, Ruokai Yin, Donghyun Lee, Arkapravo Ghosh, Priyadarshini Panda

IEEE/ACM International Conference on Computer Aided Design (ICCAD), 2026

- **Problem:** Dual-sided bit-serial accelerators skip zero bits on both operands, but each lane's cost is the product of two independently varying term counts, so lanes in a lockstep group finish at very different times, idling the faster lanes and capping PE utilization at only 56–64% in prior designs
- **Solution:** Cyclic-Balanced Pruning reshapes weights offline to equalize per-lane work, and pairwise slot donation hardware (1.2% PE area) absorbs runtime imbalance; over 90% PE utilization, up to $2.37\times$ speedup and $1.63\times$ energy efficiency across CNNs, ViTs and LLMs under iso-area constraints

NeuroFlex: Lossless Element-Level ANN–SNN Co-Execution for Efficient Sparse Inference.

Varun Manjunath, Pranav Ramesh and Gopalakrishnan Srinivasan

International Symposium on Microarchitecture (MICRO) 2026.  [Open-source artifact](#).

- **Problem:** Per-element work varies several-fold in a sparse layer, yet ANN–SNN hybrids switch per layer or tile via lossy conversion (40–45% PE utilization)
- **Solution:** Proved integer-exact ANN–SNN equivalence holds per output element, enabling lossless element-level routing via a cost-guided offline scheduler; 97–99% PE utilization, 57–67% lower EDP than an ANN-only baseline and up to $2.5\times$ speedup over an SNN-only one, with bit-identical outputs

Through Fabric: A Cross-world Thermal Covert Channel on TEE-enhanced FPGA-MPSoC Systems

Jeferson Gonzalez-Gomez, Hassan Nassar, Varun Manjunath, Lars Bauer, Joerg Henkel

Asia and South Pacific Design Automation Conference (ASP-DAC), 2025

- **Problem:** TEEs on FPGA-MPSoCs are assumed to isolate secure-world data; prior FPGA covert channels assumed bare metal and malicious hardware
- **Solution:** First thermal cross-world covert channel on a commercial FPGA-MPSoC under OP-TEE, using only a benign AES accelerator; 2 bits/s at 1.9% BER

AWARDS & HONORS

Research: DAC Young Fellow (2026) · DAC Young Fellow Best Video Presentation (2026) · MICRO Student Research Competition Finalist (2026)

Academic: DAAD WISE Scholarship, IIT Madras (Summer 2023) · Shankar Family Scholarship, IIT Madras (Fall 2022 & Fall 2024)

TALKS

BRIM: Workload-Balanced Dual-Sided Bit-Serial Sparse Inference Accelerator

ICCAD 2026 (San Jose, USA), Nov 2026

NeuroFlex: Lossless Element-Level ANN–SNN Co-Execution for Efficient Sparse Inference

MICRO 2026 (Athens, Greece), Nov 2026

TEACHING & SERVICE

TA - EE 2003, Computer Organization, Fall 2024 (Instructor: Prof. Janakiraman Viraraghavan)

Artifacts Evaluator - IEEE/ACM International Symposium on Microarchitecture (MICRO 2026)